

WILMINGTON NWS, OH, USA

WMO: 724296

Lat: 39.4214N

Lon: 83.8217W

Elev: 1067

StdP: 14.14

Time Zone: -5.00 (NAE)

Period: 98-19

WBAN: 13841

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 3.5	(c) 8.9	(d) -4.2	(e) 4.6	(f) 5.2	(g) 0.8	(h) 6.0	(i) 10.3	(j) 27.7	(k) 36.1	(l) 25.4	(m) 35.7	(n) 7.0	(o) 270	(p) 0.502

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.0	90.0	73.7	87.7	73.0	85.2	71.8	76.9	86.1	75.4	83.6	74.1	81.6	9.9	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
73.9	131.7	81.4	72.7	126.3	79.9	71.8	122.3	78.8	40.9	86.2	39.5	83.9	38.2	81.5	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 24.3	(b) 20.5	(c) 18.4	(e) -4.2	(f) 92.8	(g) 6.8	(h) 3.0	(i) -9.1	(j) 94.9	(k) -13.1	(l) 96.7	(m) -16.9	(n) 98.3	(o) -21.9	(p) 100.5		
(4) 24.3	(5) 20.5	(6) 18.4	(7) -4.3	(8) 79.2	(9) 7.0	(10) 1.7	(11) -9.3	(12) 80.5	(13) -13.4	(14) 81.4	(15) -17.3	(16) 82.4	(17) -22.4	(18) 83.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	53.0	28.9	32.1	41.3	53.3	63.4	71.0	73.6	72.3	66.4	54.7	43.4	34.1	(6)
(7)		DBStd	18.21	12.41	11.49	11.08	9.03	7.96	5.83	5.05	4.83	7.02	9.01	9.60	10.66	(7)
(8)		HDD50	2359	659	510	317	69	5	0	0	0	52	243	504		(8)
(9)		HDD65	5303	1118	922	736	360	130	17	2	5	67	339	648	959	(9)
(10)		CDD50	3447	6	8	47	168	420	631	733	690	491	198	45	10	(10)
(11)		CDD65	916	0	0	2	9	79	198	270	230	108	20	0	0	(11)
(12)		CDH74	7550	0	0	14	85	633	1598	2242	1829	987	160	2	0	(12)
(13)		CDH80	2301	0	0	1	5	129	497	767	563	308	31	0	0	(13)
(14)	Wind	WSAvg	8.4	10.5	10.1	9.8	10.0	8.2	7.3	6.3	5.8	6.3	7.9	9.0	9.5	(14)
(15)	Precipitation	PrecAvg	42.5	3.3	2.6	3.7	3.8	5.4	4.3	3.8	3.3	3.0	3.1	2.8	3.3	(15)
(16)		PrecMax	57.7	7.3	5.3	8.5	8.1	11.1	7.7	6.7	5.7	6.4	6.1	5.8	8.2	(16)
(17)		PrecMin	25.9	1.4	1.0	1.8	1.1	1.6	1.9	1.5	1.5	1.1	1.0	0.7	1.1	(17)
(18)		PrecStd	6.4	1.4	1.3	1.5	1.8	2.6	1.7	1.2	1.3	1.8	1.7	1.3	1.5	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	62.8	66.0	75.3	80.8	86.7	90.9	93.3	92.2	91.0	83.8	71.9	63.6	(19)
(20)			MCWB	58.1	59.5	60.9	64.9	70.5	72.7	76.2	73.7	70.9	68.1	60.3	58.9	(20)
(21)		2%	DB	57.2	59.3	68.6	76.7	83.2	88.1	90.0	88.4	87.2	78.5	66.3	58.5	(21)
(22)			MCWB	53.9	54.4	57.5	61.5	68.5	72.4	75.4	73.6	70.9	65.6	57.4	54.7	(22)
(23)		5%	DB	53.2	54.6	63.3	72.7	81.1	85.3	87.5	85.8	83.6	73.9	63.0	54.3	(23)
(24)			MCWB	50.4	50.0	54.4	59.9	67.6	71.1	74.0	72.5	69.6	62.6	56.3	51.1	(24)
(25)		10%	DB	47.0	49.3	59.0	68.8	77.8	82.5	84.3	83.0	80.1	69.9	58.6	49.9	(25)
(26)			MCWB	43.6	44.4	52.3	58.0	65.8	69.9	72.8	71.2	67.1	61.7	52.6	46.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.9	60.1	63.2	67.9	73.5	77.1	79.4	78.0	75.4	71.2	63.4	60.2	(27)
(28)			MCDWB	62.0	65.2	73.5	77.9	82.4	86.5	88.4	86.8	85.5	81.1	67.7	63.1	(28)
(29)		2%	WB	54.8	55.2	59.5	64.5	71.3	75.1	77.4	76.4	73.4	68.3	60.1	55.5	(29)
(30)			MCDWB	57.2	57.9	66.0	71.9	80.0	83.3	87.0	84.9	82.1	74.7	64.8	57.5	(30)
(31)		5%	WB	50.5	50.5	56.6	61.9	69.4	73.4	75.8	74.9	71.5	65.5	56.7	51.4	(31)
(32)			MCDWB	52.6	54.3	61.9	69.3	77.3	81.9	84.2	82.3	79.6	72.0	61.5	53.6	(32)
(33)		10%	WB	43.8	44.3	52.9	59.7	67.7	72.0	74.4	73.2	69.8	62.4	53.3	46.3	(33)
(34)			MCDWB	46.8	48.7	58.5	66.6	75.2	79.9	81.8	79.5	76.2	68.5	58.0	49.0	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	14.7	15.9	17.8	20.6	19.2	19.1	19.0	20.0	21.6	19.7	17.3	13.6	(35)
(36)			MCDBR	18.6	22.3	23.2	24.2	22.0	22.1	21.5	22.5	25.3	23.6	21.7	18.1	(36)
(37)			MCWBR	16.3	18.2	14.8	14.2	10.8	9.7	9.0	10.3	11.1	12.5	15.2	15.9	(37)
(38)		5% WB	MCDWB	17.3	20.8	20.2	20.7	19.1	18.5	18.9	18.5	20.4	19.7	19.0	17.4	(38)
(39)			MCWBR	16.3	18.5	14.2	13.8	10.4	9.5	9.0	9.5	10.1	11.1	15.0	16.5	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.301	0.321	0.347	0.388	0.434	0.445	0.451	0.449	0.392	0.352	0.322	0.301	(40)
(41)		taud		2.463	2.370	2.375	2.273	2.177	2.195	2.182	2.170	2.311	2.423	2.469	2.470	(41)
(42)		Ebn at Noon		277	285	288	282	269	266	263	260	269	270	266	267	(42)
(43)		Edh at Noon		25	32	35	41	47	46	46	45	37	30	25	23	(43)
(44)	All-Sky Solar Radiation	RadAvg		554	816	1137	1537	1747	1927	1861	1733	1420	1012	689	469	(44)
(45)		RadStd		51	100	97	122	146	118	107	95	136	96	82	48	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+207	+130
(47) Regional (29 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+218	+126

Nomenclature: See separate page